Taller - Consultas básicas en PostgreSQL

Asignatura: Bases de Datos I - 2026-2 **Universidad de Pamplona** - Ingeniería de Sistemas **Código estudiantil:** 1022354349 **Base de datos:** `coffee_coop_db`

Ejercicio 1. SELECT y alias - Catálogo de departamentos

a) Todas las columnas y todas las filas

```
SELECT T1.*
FROM geo.departments AS T1;
```

```
coffee_coop_db=> SELECT T1.*
coffee_coop_db-> FROM geo.departments AS T1
coffee_coop_db-> LIMIT 10;
 department_code | department_name
-----------------+-----------------
 05              | Antioquia
 08              | Atl├íntico
 11              | Bogot├í, D.C.
 13              | Bol├¡var
 15              | Boyac├í
 17              | Caldas
 18              | Caquet├í
 19              | Cauca
 20              | Cesar
 23              | C├│rdoba
(10 filas)


coffee_coop_db=>
```

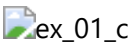
b) Código y nombre con alias

```
SELECT
    T1.department_code AS code,
    T1.department_name AS department
FROM geo.departments AS T1;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.department_code AS code,
coffee_coop_db->     T1.department_name AS department
coffee_coop_db-> FROM geo.departments AS T1
coffee_coop_db-> LIMIT 10;
 code |  department
------+--------------
 05   | Antioquia
 08   | Atl├íntico
 11   | Bogot├í, D.C.
 13   | Bol├¡var
 15   | Boyac├í
 17   | Caldas
 18   | Caquet├í
 19   | Cauca
 20   | Cesar
 23   | C├│rdoba
(10 filas)
```

c) Ordenado alfabéticamente por nombre

```sql
SELECT
    T1.department_code AS code,
    T1.department_name AS department
FROM geo.departments AS T1
ORDER BY department;
```



Ejercicio 2. WHERE con booleanos - Variedades de café

a) Variedades resistentes a la roya

```sql
SELECT
    T1.variety_name AS variety,
    T1.species AS species,
    T1.description AS description
FROM farm.coffee_varieties AS T1
WHERE T1.is_rust_resistant = TRUE;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.variety_name AS variety,
coffee_coop_db->     T1.species AS species,
coffee_coop_db->     T1.description AS description
coffee_coop_db-> FROM farm.coffee_varieties AS T1
coffee_coop_db-> WHERE T1.is_rust_resistant = TRUE
coffee_coop_db-> LIMIT 10;
  variety   | species |                                  description

------------+---------+-------------------------------------------------------
----------------
 Castillo   | arabica | Variedad colombiana resistente a la roya, la m├ís se
mbrada del pa├¡s.
 Colombia   | arabica | Variedad compuesta resistente a la roya, antecesora
de Castillo.
 Cenicaf├® 1 | arabica | Variedad de porte bajo con resistencia a la roya.
 Tabi       | arabica | Variedad de porte alto resistente a la roya.
 Robusta    | robusta | Especie Coffea canephora, de mayor contenido de cafe
├¡na.
(5 filas)
```

b) Arabica no resistentes, nombre descendente

```sql
SELECT
    T1.variety_name AS variety,
    T1.species AS species,
    T1.description AS description
FROM farm.coffee_varieties AS T1
WHERE T1.is_rust_resistant = FALSE
  AND T1.species = 'arabica'
ORDER BY variety DESC;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.variety_name AS variety,
coffee_coop_db->     T1.species AS species,
coffee_coop_db->     T1.description AS description
coffee_coop_db-> FROM farm.coffee_varieties AS T1
coffee_coop_db-> WHERE T1.is_rust_resistant = FALSE
coffee_coop_db->   AND T1.species = 'arabica'
coffee_coop_db-> ORDER BY variety DESC
coffee_coop_db-> LIMIT 10;
    variety    | species |                        description

---------------+---------+-----------------------------------------------------
------
 T├¡pica        | arabica | Variedad tradicional de porte alto.
 Pacamara      | arabica | Cruce de Pacas y Maragogipe, grano grande.
 Maragogipe    | arabica | Variedad de grano grande.
 Geisha        | arabica | Variedad de especialidad con perfil floral.
 Caturra       | arabica | Mutaci├│n de Borb├│n de porte bajo y alta product
ividad.
 Borb├│n Rosado | arabica | Variedad de especialidad de fruto rosado.
 Borb├│n        | arabica | Variedad tradicional de buena calidad en taza.
(7 filas)
```

Ejercicio 3. Columnas calculadas - Primas de certificación

a) Certificación, emisor y prima por kg

```
SELECT
    T1.certification_name AS certification,
    T1.issuing_body AS issuer,
    T1.premium_per_kg AS premium_per_kg_cop
FROM farm.certifications AS T1
ORDER BY premium_per_kg_cop DESC;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.certification_name AS certification,
coffee_coop_db->     T1.issuing_body AS issuer,
coffee_coop_db->     T1.premium_per_kg AS premium_per_kg_cop
coffee_coop_db-> FROM farm.certifications AS T1
coffee_coop_db-> ORDER BY premium_per_kg_cop DESC
coffee_coop_db-> LIMIT 10;
    certification    |              issuer               | premium_per_kg_cop
---------------------+-----------------------------------+--------------------
 Org├ínico           | Ente certificador org├ínico       |             800.00
 Bird Friendly       | Smithsonian Migratory Bird Center |             500.00
 Fairtrade           | Fairtrade International           |             400.00
 Nespresso AAA       | Nespresso / Rainforest Alliance   |             350.00
 Rainforest Alliance | Rainforest Alliance               |             300.00
 C.A.F.E. Practices  | Starbucks / SCS Global Services   |             250.00
 4C                  | 4C Services                       |             150.00
(7 filas)
```

b) Columna calculada: prima por carga (premium_per_kg * 125)

```
SELECT
    T1.certification_name AS certification,
    T1.issuing_body AS issuer,
    T1.premium_per_kg AS premium_per_kg_cop,
    T1.premium_per_kg * 125 AS premium_per_load_cop
FROM farm.certifications AS T1
ORDER BY premium_per_kg_cop DESC;
```

```
coffee_coop_db=> chcp 65001
coffee_coop_db-> $env:PGCLIENTENCODING = "UTF8"
coffee_coop_db-> psql -U tu_usuario -d coffee_coop_db
coffee_coop_db-> SELECT
coffee_coop_db->     T1.certification_name AS certification,
coffee_coop_db->     T1.issuing_body AS issuer,
coffee_coop_db->     T1.premium_per_kg AS premium_per_kg_cop,
coffee_coop_db->     T1.premium_per_kg * 125 AS premium_per_load_cop
coffee_coop_db-> FROM farm.certifications AS T1
coffee_coop_db-> ORDER BY premium_per_kg_cop DESC
coffee_coop_db-> LIMIT 10;
ERROR:  error de sintaxis en o cerca de Â«chcpÂ»
LÍNEA 1: chcp 65001
         ^
coffee_coop_db=> SELECT
coffee_coop_db->     T1.certification_name AS certification,
coffee_coop_db->     T1.issuing_body AS issuer,
coffee_coop_db->     T1.premium_per_kg AS premium_per_kg_cop,
coffee_coop_db->     T1.premium_per_kg * 125 AS premium_per_load_cop
coffee_coop_db-> FROM farm.certifications AS T1
coffee_coop_db-> ORDER BY premium_per_kg_cop DESC
coffee_coop_db-> LIMIT 10;
    certification    |              issuer               | premium_per_kg_cop | premium_per_load_cop
---------------------+-----------------------------------+--------------------+----------------------
 Org├ínico           | Ente certificador org├ínico       |             800.00 |            100000.00
 Bird Friendly       | Smithsonian Migratory Bird Center |             500.00 |             62500.00
 Fairtrade           | Fairtrade International           |             400.00 |             50000.00
 Nespresso AAA       | Nespresso / Rainforest Alliance   |             350.00 |             43750.00
 Rainforest Alliance | Rainforest Alliance               |             300.00 |             37500.00
 C.A.F.E. Practices  | Starbucks / SCS Global Services   |             250.00 |             31250.00
 4C                  | 4C Services                       |             150.00 |             18750.00
(7 filas)
```

c) Prima por kg mayor o igual a 350 COP

```sql
SELECT
    T1.certification_name AS certification,
    T1.issuing_body AS issuer,
    T1.premium_per_kg AS premium_per_kg_cop
FROM farm.certifications AS T1
WHERE T1.premium_per_kg >= 350
ORDER BY premium_per_kg_cop DESC;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.certification_name AS certification,
coffee_coop_db->     T1.issuing_body AS issuer,
coffee_coop_db->     T1.premium_per_kg AS premium_per_kg_cop
coffee_coop_db-> FROM farm.certifications AS T1
coffee_coop_db-> WHERE T1.premium_per_kg >= 350
coffee_coop_db-> ORDER BY premium_per_kg_cop DESC
coffee_coop_db-> LIMIT 10;
 certification |              issuer               | premium_per_kg_cop
---------------+-----------------------------------+--------------------
 Org├ínico      | Ente certificador org├ínico       |             800.00
 Bird Friendly | Smithsonian Migratory Bird Center |             500.00
 Fairtrade     | Fairtrade International           |             400.00
 Nespresso AAA | Nespresso / Rainforest Alliance   |             350.00
(4 filas)
```

Ejercicio 4. WHERE con LIKE - Municipios de Norte de Santander

a) Municipios de Norte de Santander (código 54)

```sql
SELECT
    T1.municipality_code AS municipality_code,
    T1.municipality_name AS municipality
FROM geo.municipalities AS T1
WHERE T1.municipality_code LIKE '54%'
ORDER BY municipality;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.municipality_code AS municipality_code,
coffee_coop_db->     T1.municipality_name AS municipality
coffee_coop_db-> FROM geo.municipalities AS T1
coffee_coop_db-> WHERE T1.municipality_code LIKE '54%'
coffee_coop_db-> ORDER BY municipality
coffee_coop_db-> LIMIT 10;
 municipality_code | municipality
-------------------+--------------
 54003             | ├übrego
 54051             | Arboledas
 54099             | Bochalema
 54109             | Bucarasica
 54128             | C├ích
 54125             | C├ícota
 54172             | Chin├ícota
 54174             | Chitag├í
 54206             | Convenci├│n
 54001             | C├│cuta
(10 filas)
```

b) Nombre que comienza con la palabra San

```sql
SELECT
    T1.municipality_code AS municipality_code,
    T1.municipality_name AS municipality
FROM geo.municipalities AS T1
WHERE T1.municipality_code LIKE '54%'
  AND T1.municipality_name LIKE 'San %'
ORDER BY municipality;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.municipality_code AS municipality_code,
coffee_coop_db->     T1.municipality_name AS municipality
coffee_coop_db-> FROM geo.municipalities AS T1
coffee_coop_db-> WHERE T1.municipality_code LIKE '54%'
coffee_coop_db->   AND T1.municipality_name LIKE 'San %'
coffee_coop_db-> ORDER BY municipality
coffee_coop_db-> LIMIT 10;
 municipality_code | municipality
-------------------+--------------
 54670             | San Calixto
 54673             | San Cayetano
(2 filas)
```

c) Nombre que contiene Pamplona (ILIKE)

```sql
SELECT
    T1.municipality_code AS municipality_code,
    T1.municipality_name AS municipality
FROM geo.municipalities AS T1
WHERE T1.municipality_name ILIKE '%pamplona%'
ORDER BY municipality;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.municipality_code AS municipality_code,
coffee_coop_db->     T1.municipality_name AS municipality
coffee_coop_db-> FROM geo.municipalities AS T1
coffee_coop_db-> WHERE T1.municipality_name ILIKE '%pamplona%'
coffee_coop_db-> ORDER BY municipality
coffee_coop_db-> LIMIT 10;
 municipality_code | municipality
-------------------+--------------
 54518             | Pamplona
(1 fila)
```

Ejercicio 5. IS NULL y operadores lógicos - Puntos de compra

a) Puntos de compra inactivos

```sql
SELECT
    T1.point_name AS point,
    T1.address_line AS address,
    T1.municipality_code AS municipality_code
FROM buy.collection_points AS T1
WHERE T1.is_active = FALSE;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.point_name AS point,
coffee_coop_db->     T1.address_line AS address,
coffee_coop_db->     T1.municipality_code AS municipality_code
coffee_coop_db-> FROM buy.collection_points AS T1
coffee_coop_db-> WHERE T1.is_active = FALSE
coffee_coop_db-> LIMIT 10;
           point            |      address       | municipality_code
----------------------------+--------------------+-------------------
 Punto de Compra El Banco 1 | Calle 17 # 34-54   | 47245
 Punto de Compra Popay├ín 1 | Carrera 21 # 4-95  | 19001
 Punto de Compra Ap├¡a 2     | Calle 9 # 14-56    | 66045
 Punto de Compra Tello 1    | Calle 8 # 13-23    | 41799
 Punto de Compra Turbo 1    | Avenida 32 # 34-61 | 05837
 Punto de Compra Paicol 2   | Calle 2 # 30-84    | 41518
 Punto de Compra Rionegro 1 | Carrera 31 # 59-80 | 68615
 Punto de Compra Ulloa 1    | Calle 15 # 3-50    | 76845
 Punto de Compra N├ítaga 1   | Diagonal 30 # 7-59 | 41483
(9 filas)
```

b) Puntos sin teléfono, ordenados por nombre

```sql
SELECT
    T1.collection_point_id,
    T1.point_name,
    T1.municipality_code,
    T1.address_line,
    T1.phone_number,
    T1.is_active
FROM buy.collection_points AS T1
WHERE T1.phone_number IS NULL
ORDER BY T1.point_name;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.collection_point_id,
coffee_coop_db->     T1.point_name,
coffee_coop_db->     T1.municipality_code,
coffee_coop_db->     T1.address_line,
coffee_coop_db->     T1.phone_number,
coffee_coop_db->     T1.is_active
coffee_coop_db-> FROM buy.collection_points AS T1
coffee_coop_db-> WHERE T1.phone_number IS NULL
coffee_coop_db-> ORDER BY T1.point_name
coffee_coop_db-> LIMIT 10;
 collection_point_id |         point_name          | municipality_code |      address_line      | phone_number | is_active
---------------------+-----------------------------+-------------------+------------------------+--------------+-----------
                  60 | Punto de Compra Aipe 1      | 41016             | Carrera 13 # 17-16     |              | t
                  54 | Punto de Compra Anserma 1   | 17042             | Calle 39 # 7-96        |              | t
                  57 | Punto de Compra Ap├¡a 2      | 66045             | Calle 9 # 14-56        |              | f
                 162 | Punto de Compra Baraya 1    | 41078             | Calle 1 # 19-77        |              | t
                 163 | Punto de Compra Bochalema 1 | 54099             | Carrera 25 # 47-89     |              | t
                  15 | Punto de Compra Bosconia 1  | 20060             | Carrera 26 # 26-43     |              | t
                  20 | Punto de Compra Cajib├¡o 1  | 19130             | Carrera 15 # 32-45     |              | t
                 146 | Punto de Compra Cisneros 1  | 05190             | Calle 37 # 10-5        |              | t
                 176 | Punto de Compra Coello 1    | 73200             | Transversal 20 # 49-66 |              | t
                 152 | Punto de Compra El Carmen 2 | 54245             | Calle 3 # 34-88        |              | t
(10 filas)
```

c) Inactivos o sin teléfono

```sql
SELECT
    T1.point_name,
```

```
    T1.address_line,
    T1.municipality_code,
    T1.phone_number,
    T1.is_active
FROM buy.collection_points AS T1
WHERE T1.is_active = FALSE
   OR T1.phone_number IS NULL;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.point_name,
coffee_coop_db->     T1.address_line,
coffee_coop_db->     T1.municipality_code,
coffee_coop_db->     T1.phone_number,
coffee_coop_db->     T1.is_active
coffee_coop_db-> FROM buy.collection_points AS T1
coffee_coop_db-> WHERE T1.is_active = FALSE
coffee_coop_db->    OR T1.phone_number IS NULL
coffee_coop_db-> LIMIT 10;
         point_name          |      address_line      | municipality_code | phone_number | is_active
-----------------------------+------------------------+-------------------+--------------+-----------
 Punto de Compra El Banco 1  | Calle 17 # 34-54       | 47245             | 6055026534   | f
 Punto de Compra Popay├ín 1  | Carrera 21 # 4-95      | 19001             | 6057341141   | f
 Punto de Compra Hatonuevo 1 | Calle 7 # 16-89        | 44378             |              | t
 Punto de Compra Bosconia 1  | Carrera 26 # 26-43     | 20060             |              | t
 Punto de Compra Cajib├¡o 1  | Carrera 15 # 32-45     | 19130             |              | t
 Punto de Compra Anserma 1   | Calle 39 # 7-96        | 17042             |              | t
 Punto de Compra Ap├¡a 2     | Calle 9 # 14-56        | 66045             |              | f
 Punto de Compra Purac├® 1   | Calle 16 # 50-5        | 19585             |              | t
 Punto de Compra Aipe 1      | Carrera 13 # 17-16     | 41016             |              | t
 Punto de Compra Marsella 1  | Transversal 19 # 42-92 | 66440             |              | t
(10 filas)
```

Ejercicio 6. Filtros combinados - Asociados de Pamplona

a) Asociadas (gender = 'F') del municipio 54518

```
SELECT
    T1.document_number AS document,
    T1.first_name AS first_name,
    T1.last_name AS last_name,
    T1.phone_number AS phone
FROM farm.farmers AS T1
WHERE T1.gender = 'F'
  AND T1.municipality_code = '54518'
ORDER BY last_name, first_name;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.document_number AS document,
coffee_coop_db->     T1.first_name AS first_name,
coffee_coop_db->     T1.last_name AS last_name,
coffee_coop_db->     T1.phone_number AS phone
coffee_coop_db-> FROM farm.farmers AS T1
coffee_coop_db-> WHERE T1.gender = 'F'
coffee_coop_db->   AND T1.municipality_code = '54518'
coffee_coop_db-> ORDER BY last_name, first_name
coffee_coop_db-> LIMIT 10;
  document   |   first_name    |    last_name     |   phone
-------------+-----------------+------------------+------------
 768461420   | Marleny         | Agudelo Sandoval | 3242280898
 931079279   | Edilma          | ├ülvarez Bonilla  | 3236902029
 463842401   | Roc├¡o          | ├ülvarez C├írdenas | 3047898372
 788628125   | M├│nica         | ├ülvarez Carrillo | 3059237130
 939635303   | Marleny         | ├ülvarez Correa   | 3232410671
 1089620113  | Myriam Yuliana  | Arango Narv├íez   | 3178267449
 172555225   | Edilma Gabriela | Arias ├ülvarez    | 3213658326
 550696749   | Viviana         | Arias Ram├¡rez    | 3204850778
 797017145   | Isabella        | Barrera Cabrera  | 3041744702
 659942505   | Mariana         | Barrera Correa   | 3126485460
(10 filas)
```

b) Asociados del municipio 54518 sin correo

```sql
SELECT
    T1.farmer_id,
    T1.document_number,
    T1.first_name,
    T1.last_name,
    T1.birth_date,
    T1.gender,
    T1.phone_number,
    T1.email,
    T1.membership_date,
    T1.is_active,
    T1.municipality_code
FROM farm.farmers AS T1
WHERE T1.municipality_code = '54518'
  AND T1.email IS NULL;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.farmer_id, T1.document_number, T1.first_name, T1.last_name,
coffee_coop_db->     T1.birth_date, T1.gender, T1.phone_number, T1.email,
coffee_coop_db->     T1.membership_date, T1.is_active, T1.municipality_code
coffee_coop_db-> FROM farm.farmers AS T1
coffee_coop_db-> WHERE T1.municipality_code = '54518'
coffee_coop_db->   AND T1.email IS NULL
coffee_coop_db-> LIMIT 10;
 farmer_id | document_number |    first_name    |    last_name     | birth_date | gender | phone_number | email | membership_date | is_active | municipality_code
-----------+-----------------+------------------+------------------+------------+--------+--------------+-------+-----------------+-----------+-------------------
       775 | 712758473       | Wilson           | L├│pez Villamizar | 1996-04-26 | M      | 3055094275   |       | 2016-02-19      | t         | 54518
      2716 | 727945493       | Gloria           | Mar├¡n Rivera     | 1975-06-11 | O      | 3225571567   |       | 2021-01-28      | t         | 54518
      3739 | 549054771       | Mar├¡a            | Burbano Duarte   | 1964-08-24 | F      | 3183590387   |       | 2004-09-23      | t         | 54518
      7898 | 24999630        | Esperanza        | Romero Betancur  | 1955-08-11 | F      | 3029623613   |       | 2015-08-30      | t         | 54518
      8485 | 914044314       | Anc├¡zar Cristian | Polan├¡a Delgado  | 1997-04-18 | M      | 3028825809   |       | 2021-02-24      | f         | 54518
     10999 | 343256743       | Dora             | Hern├índez Correa | 1952-12-18 | F      | 3166719225   |       | 2014-03-08      | t         | 54518
     12711 | 1148351420      | In├®s            | Jim├®nez Jaimes   | 1966-09-14 | F      | 3104090231   |       | 2008-02-11      | t         | 54518
     14977 | 1117597835      | Cecilia Lorena   | D├¡az Garc├¡a     | 1953-09-22 | F      | 3207660684   |       | 2017-01-11      | f         | 54518
     20108 | 749444168       | Emilio           | Delgado Molina   | 1992-02-05 | M      | 3118015821   |       | 2016-03-11      | t         | 54518
     20979 | 940051048       | David Germ├ín     | Romero Bonilla   | 1980-09-08 | M      | 3018117176   |       | 2002-10-11      | t         | 54518
(10 filas)
```

c) Asociados inactivos, del más antiguo al más reciente

```sql
SELECT
    T1.farmer_id,
    T1.document_number,
    T1.first_name,
    T1.last_name,
    T1.birth_date,
    T1.gender,
    T1.phone_number,
    T1.email,
    T1.membership_date,
    T1.is_active,
    T1.municipality_code
FROM farm.farmers AS T1
WHERE T1.municipality_code = '54518'
  AND T1.is_active = FALSE
ORDER BY T1.membership_date ASC;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.farmer_id, T1.document_number, T1.first_name, T1.last_name,
coffee_coop_db->     T1.birth_date, T1.gender, T1.phone_number, T1.email,
coffee_coop_db->     T1.membership_date, T1.is_active, T1.municipality_code
coffee_coop_db-> FROM farm.farmers AS T1
coffee_coop_db-> WHERE T1.municipality_code = '54518'
coffee_coop_db->   AND T1.is_active = FALSE
coffee_coop_db-> ORDER BY T1.membership_date ASC
coffee_coop_db-> LIMIT 10;
 farmer_id | document_number |   first_name   |     last_name      | birth_date | gender | phone_number |             email              | membership_date | is_active | municipality_code
-----------+-----------------+----------------+--------------------+------------+--------+--------------+--------------------------------+-----------------+-----------+-------------------
     99920 | 230971064       | Manuel         | Agudelo Jim├®nez    | 1970-06-03 | M      | 3219718862   | manuel.agudelo99920@hotmail.com | 1997-03-29      | f         | 54518
    111416 | 534023693       | Myriam         | Valencia Moreno    | 1962-04-04 | F      | 3241389451   | myriam.valencia111416@gmail.com | 1998-11-03      | f         | 54518
     53823 | 33787285        | Rub├®n         | Ramos Chamorro     | 1978-12-05 | M      | 3244273375   |                                | 1999-03-02      | f         | 54518
     42933 | 1135293825      | Abelardo       | P├®rez Castro       | 1976-12-13 | M      | 3245612074   |                                | 2003-06-19      | f         | 54518
     94576 | 478755763       | Nicol├ís        | Rivera Duque       | 1952-03-06 | M      | 3029310183   | nicolas.rivera94576@gmail.com  | 2003-08-11      | f         | 54518
    111147 | 160538872       | Liliana        | Mosquera Uribe     | 1969-12-23 | F      | 3119365364   |                                | 2007-09-27      | f         | 54518
     22657 | 223974138       | Rodrigo Nelson | Cabrera Villamizar | 1978-02-28 | M      | 3502554241   | rodrigo.cabrera22657@outlook.com | 2009-05-02      | f         | 54518
     25949 | 10424994        | Cristian       | Herrera Zuluaga    | 1988-03-24 | M      | 3242295228   | cristian.herrera25949@yahoo.com | 2010-04-28      | f         | 54518
    113393 | 1015411133      | Santiago       | Berm├║dez Ruiz      | 1977-08-22 | M      | 3003993000   |                                | 2010-11-30      | f         | 54518
     66985 | 1090968710      | Ruth           | Rodr├¡guez P├®rez   | 1992-04-13 | F      | 3021808053   |                                | 2010-12-11      | f         | 54518
(10 filas)
```

Ejercicio 7. BETWEEN y ORDER BY múltiple - Fincas de altura

a) Fincas sobre 1800 m.s.n.m.

```sql
SELECT
    T1.farm_name AS farm,
    T1.rural_district AS rural_district,
    T1.total_area_ha AS area_ha,
    T1.altitude_masl AS altitude_masl
FROM farm.farms AS T1
WHERE T1.municipality_code = '54518'
  AND T1.altitude_masl > 1800
ORDER BY altitude_masl DESC;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.farm_name AS farm,
coffee_coop_db->     T1.rural_district AS rural_district,
coffee_coop_db->     T1.total_area_ha AS area_ha,
coffee_coop_db->     T1.altitude_masl AS altitude_masl
coffee_coop_db-> FROM farm.farms AS T1
coffee_coop_db-> WHERE T1.municipality_code = '54518'
coffee_coop_db->   AND T1.altitude_masl > 1800
coffee_coop_db-> ORDER BY altitude_masl DESC
coffee_coop_db-> LIMIT 10;
     farm      | rural_district | area_ha | altitude_masl
---------------+----------------+---------+---------------
 El Guayabo    | El Rosal       |    1.21 |          2300
 La Cascada    | Caima          |    2.49 |          2300
 Santa Clara   | La Legua       |    2.62 |          2300
 El Vergel     | Sabagua        |    1.11 |          2300
 La Isabela    | Garc┼¡a        |    1.36 |          2275
 El Refugio    | Monteadentro   |    4.89 |          2242
 Cafetal Viejo | Caima          |    5.29 |          2229
 San Jos┼®     | Alcaparral     |    3.12 |          2203
 El Refugio    | Sabaneta Baja  |    1.93 |          2195
 Villa Hermosa | Pe┼as          |    0.30 |          2181
(10 filas)
```

b) Fincas entre 1600 y 1800 m.s.n.m.

```sql
SELECT
    T1.farm_id,
    T1.farmer_id,
    T1.municipality_code,
    T1.farm_name,
    T1.rural_district,
    T1.total_area_ha,
    T1.altitude_masl,
    T1.latitude,
    T1.longitude,
    T1.registered_at
FROM farm.farms AS T1
WHERE T1.municipality_code = '54518'
  AND T1.altitude_masl BETWEEN 1600 AND 1800
ORDER BY T1.rural_district ASC, T1.total_area_ha DESC;
```

```
coffee_coop_db=> SELECT T1.farm_id, T1.farmer_id, T1.municipality_code, T1.farm_name, T1.rural_district, T1.total_area_ha, T1.altitude_masl, T1.latitude, T1.longitude, T1.registered_at
coffee_coop_db-> FROM farm.farms AS T1
coffee_coop_db-> WHERE T1.municipality_code = '54518'
coffee_coop_db->   AND T1.altitude_masl BETWEEN 1600 AND 1800
coffee_coop_db-> ORDER BY T1.rural_district ASC, T1.total_area_ha DESC
coffee_coop_db-> LIMIT 10;
 farm_id | farmer_id | municipality_code |  farm_name   |   rural_district   | total_area_ha | altitude_masl | latitude | longitude |    registered_at
---------+-----------+-------------------+--------------+--------------------+---------------+---------------+----------+-----------+---------------------
  123713 |     79656 | 54518             | Los Pinos    | Alcaparral         |          1.67 |          1771 |          |           | 2017-01-16 15:53:24
  135683 |     87442 | 54518             | Miraflores   | Alcaparral         |          1.29 |          1636 |          |           | 2021-12-26 07:45:34
  220768 |    142523 | 54518             | Monteverde   | Alto Grande        |          2.59 |          1794 | 8.270950 | -72.601354 | 2011-05-25 12:18:39
  106381 |     68437 | 54518             | La Loma      | Alto Grande        |          2.09 |          1667 | 7.760726 | -73.032128 | 2021-08-23 13:04:30
  162550 |    104763 | 54518             | Bellavista   | Alto Grande        |          1.80 |          1704 |          |           | 2017-01-20 07:52:05
  172543 |    111220 | 54518             | El Refugio   | Alto Grande        |          1.23 |          1661 | 7.945377 | -72.671696 | 1999-03-05 16:30:30
  210481 |    135840 | 54518             | El Jard┼¡n   | Cabecera Municipal |          3.26 |          1620 | 8.219683 | -73.151808 | 2015-04-21 08:19:41
   95792 |     61650 | 54518             | El Ed┼®n     | Cabecera Municipal |          2.89 |          1603 |          |           | 2006-05-21 15:32:21
  204909 |    132229 | 54518             | La Milagrosa | Cabecera Municipal |          2.65 |          1643 | 7.673836 | -72.492972 | 2020-04-02 07:52:05
    5849 |      3739 | 54518             | La Divisa    | Cabecera Municipal |          1.32 |          1765 | 7.322561 | -72.767022 | 2006-05-26 14:29:53
(10 filas)
```

c) Fincas sin coordenadas

```sql
SELECT
    T1.farm_id,
    T1.farmer_id,
    T1.municipality_code,
    T1.farm_name,
    T1.rural_district,
    T1.total_area_ha,
    T1.altitude_masl,
    T1.latitude,
    T1.longitude,
    T1.registered_at
FROM farm.farms AS T1
WHERE T1.municipality_code = '54518'
  AND (T1.latitude IS NULL OR T1.longitude IS NULL);
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.farm_id, T1.farmer_id, T1.municipality_code, T1.farm_name,
coffee_coop_db->     T1.rural_district, T1.total_area_ha, T1.altitude_masl,
coffee_coop_db->     T1.latitude, T1.longitude, T1.registered_at
coffee_coop_db-> FROM farm.farms AS T1
coffee_coop_db-> WHERE T1.municipality_code = '54518'
coffee_coop_db->   AND (T1.latitude IS NULL OR T1.longitude IS NULL)
coffee_coop_db-> LIMIT 10;
 farm_id | farmer_id | municipality_code |   farm_name   |   rural_district   | total_area_ha | altitude_masl | latitude | longitude |    registered_at
---------+-----------+-------------------+---------------+--------------------+---------------+---------------+----------+-----------+---------------------
    3904 |      2501 | 54518             | El Recuerdo   | Alto Grande        |          1.47 |          1875 |          |           | 2017-11-29 08:20:29
    4243 |      2716 | 54518             | La Esperanza  | El Palchal         |          1.64 |          1345 |          |           | 2021-02-08 17:28:44
    7001 |      4481 | 54518             | Las Delicias  | Caima              |          1.01 |          1308 |          |           | 2018-11-22 14:58:23
    8213 |      5280 | 54518             | Los Arrayanes | Cabecera Municipal |          3.66 |          1452 |          |           | 2017-12-03 11:26:22
    9863 |      6320 | 54518             | El Brillante  | Pe├▒as             |          2.06 |          1274 |          |           | 2020-11-15 09:28:06
   11687 |      7509 | 54518             | El Porvenir   | Fontib├│n          |          4.75 |          1418 |          |           | 2014-03-09 09:33:03
   12284 |      7898 | 54518             | La Florida    | Pe├▒as             |          1.16 |          1657 |          |           | 2021-01-29 09:33:26
   13216 |      8485 | 54518             | Los Arrayanes | Navarro            |          1.42 |          1101 |          |           | 2021-09-28 09:53:11
   15110 |      9714 | 54518             | El Guayabo    | San Francisco      |          3.20 |          1718 |          |           | 2012-07-21 11:24:18
   17013 |     10921 | 54518             | El Encanto    | Fontib├│n          |          0.75 |          1639 |          |           | 2019-06-09 07:43:23
(10 filas)
```

Ejercicio 8. IN y columnas calculadas - Entregas del día

a) Entregas del 30 de septiembre de 2026

```sql
SELECT
    T1.delivery_id AS delivery,
    T1.farm_id AS farm_id,
    T1.collection_point_id AS collection_point_id,
    T1.delivered_at AS received_at,
    T1.coffee_type AS coffee_type,
    T1.gross_weight_kg AS gross_weight_kg
FROM buy.deliveries AS T1
WHERE T1.delivery_id LIKE 'DLV-20260930-%'
ORDER BY received_at;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.delivery_id AS delivery,
coffee_coop_db->     T1.farm_id AS farm_id,
coffee_coop_db->     T1.collection_point_id AS collection_point_id,
coffee_coop_db->     T1.delivered_at AS received_at,
coffee_coop_db->     T1.coffee_type AS coffee_type,
coffee_coop_db->     T1.gross_weight_kg AS gross_weight_kg
coffee_coop_db-> FROM buy.deliveries AS T1
coffee_coop_db-> WHERE T1.delivery_id LIKE 'DLV-20260930-%'
coffee_coop_db-> ORDER BY received_at
coffee_coop_db-> LIMIT 10;
      delivery       | farm_id | collection_point_id |     received_at     |  coffee_type   | gross_weight_kg
---------------------+---------+---------------------+---------------------+----------------+-----------------
 DLV-20260930-000001 |  121925 |                  68 | 2026-09-30 06:01:10 | dry_parchment  |           51.53
 DLV-20260930-000002 |   39639 |                  48 | 2026-09-30 06:03:32 | dry_parchment  |          124.98
 DLV-20260930-000003 |  228920 |                  80 | 2026-09-30 06:04:40 | wet_parchment  |          112.33
 DLV-20260930-000004 |    2649 |                  90 | 2026-09-30 06:07:26 | dry_parchment  |          138.29
 DLV-20260930-000005 |  163723 |                  10 | 2026-09-30 06:07:51 | dry_parchment  |          256.42
 DLV-20260930-000006 |   58408 |                 170 | 2026-09-30 06:11:03 | dry_parchment  |          139.81
 DLV-20260930-000007 |  201601 |                  57 | 2026-09-30 06:15:25 | dry_parchment  |          404.97
 DLV-20260930-000008 |  135490 |                  53 | 2026-09-30 06:16:53 | dry_parchment  |          175.92
 DLV-20260930-000009 |  108211 |                  54 | 2026-09-30 06:17:34 | dry_parchment  |           56.80
 DLV-20260930-000010 |   37288 |                  67 | 2026-09-30 06:20:30 | dry_parchment  |          269.43
(10 filas)
```

b) Entregas rechazadas con peso promedio por bulto

```sql
SELECT
    T1.delivery_id,
    T1.farm_id,
    T1.collection_point_id,
    T1.delivered_at,
    T1.coffee_type,
    T1.bag_count,
    T1.gross_weight_kg,
    T1.status,
    T1.gross_weight_kg / T1.bag_count AS avg_kg_per_bag
FROM buy.deliveries AS T1
WHERE T1.delivery_id LIKE 'DLV-20260930-%'
  AND T1.status = 'rejected';
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.delivery_id, T1.farm_id, T1.collection_point_id, T1.delivered_at,
coffee_coop_db->     T1.coffee_type, T1.bag_count, T1.gross_weight_kg, T1.status,
coffee_coop_db->     T1.gross_weight_kg / T1.bag_count AS avg_kg_per_bag
coffee_coop_db-> FROM buy.deliveries AS T1
coffee_coop_db-> WHERE T1.delivery_id LIKE 'DLV-20260930-%'
coffee_coop_db->   AND T1.status = 'rejected'
coffee_coop_db-> LIMIT 10;
     delivery_id     | farm_id | collection_point_id |    delivered_at     |  coffee_type   | bag_count | gross_weight_kg |  status  |   avg_kg_per_bag
---------------------+---------+---------------------+---------------------+----------------+-----------+-----------------+----------+--------------------
 DLV-20260930-000253 |  143807 |                 102 | 2026-09-30 15:05:34 | dry_parchment  |         5 |          253.15 | rejected | 50.6300000000000000
 DLV-20260930-000006 |   58408 |                 170 | 2026-09-30 06:11:03 | dry_parchment  |         3 |          139.81 | rejected | 46.6033333333333333
 DLV-20260930-000070 |   69157 |                   3 | 2026-09-30 08:19:57 | dry_parchment  |         3 |          147.13 | rejected | 49.0433333333333333
 DLV-20260930-000165 |  107038 |                 162 | 2026-09-30 12:15:01 | dry_parchment  |         6 |          348.59 | rejected | 58.0983333333333333
 DLV-20260930-000214 |  223116 |                  44 | 2026-09-30 13:53:56 | dry_parchment  |         3 |          177.63 | rejected | 59.2100000000000000
 DLV-20260930-000223 |  216415 |                  45 | 2026-09-30 14:19:21 | dry_parchment  |         7 |          338.32 | rejected | 48.3314285714285714
 DLV-20260930-000083 |   66435 |                   2 | 2026-09-30 09:20:30 | dry_parchment  |         2 |           89.95 | rejected | 44.9750000000000000
 DLV-20260930-000109 |  216626 |                   6 | 2026-09-30 10:04:15 | dry_parchment  |         5 |          311.88 | rejected | 62.3760000000000000
 DLV-20260930-000147 |   77310 |                 103 | 2026-09-30 11:17:45 | wet_parchment  |         1 |           66.42 | rejected | 66.4200000000000000
(9 filas)
```

c) Café wet_parchment o cherry (IN)

```sql
SELECT
    T1.delivery_id,
    T1.farm_id,
    T1.collection_point_id,
    T1.delivered_at,
    T1.coffee_type,
```

```
    T1.bag_count,
    T1.gross_weight_kg,
    T1.status
FROM buy.deliveries AS T1
WHERE T1.delivery_id LIKE 'DLV-20260930-%'
  AND T1.coffee_type IN ('wet_parchment', 'cherry')
ORDER BY T1.gross_weight_kg DESC;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.delivery_id, T1.farm_id, T1.collection_point_id, T1.delivered_at,
coffee_coop_db->     T1.coffee_type, T1.bag_count, T1.gross_weight_kg, T1.status
coffee_coop_db-> FROM buy.deliveries AS T1
coffee_coop_db-> WHERE T1.delivery_id LIKE 'DLV-20260930-%'
coffee_coop_db->   AND T1.coffee_type IN ('wet_parchment', 'cherry')
coffee_coop_db-> ORDER BY T1.gross_weight_kg DESC
coffee_coop_db-> LIMIT 10;
     delivery_id     | farm_id | collection_point_id |    delivered_at     |  coffee_type  | bag_count | gross_weight_kg |  status
---------------------+---------+---------------------+---------------------+---------------+-----------+-----------------+----------
 DLV-20260930-000081 |   19338 |                 117 | 2026-09-30 09:17:37 | cherry        |         9 |          475.13 | received
 DLV-20260930-000187 |  231839 |                 159 | 2026-09-30 12:47:52 | cherry        |         6 |          339.30 | received
 DLV-20260930-000105 |   13478 |                  37 | 2026-09-30 09:57:09 | wet_parchment |         5 |          277.63 | settled
 DLV-20260930-000028 |  182397 |                 133 | 2026-09-30 07:13:46 | wet_parchment |         4 |          258.00 | received
 DLV-20260930-000133 |   44726 |                  59 | 2026-09-30 10:51:03 | cherry        |         4 |          254.83 | received
 DLV-20260930-000094 |   99627 |                 133 | 2026-09-30 09:31:20 | cherry        |         4 |          246.96 | received
 DLV-20260930-000129 |   89312 |                  76 | 2026-09-30 10:44:39 | wet_parchment |         4 |          206.71 | received
 DLV-20260930-000161 |  101318 |                  24 | 2026-09-30 12:06:51 | wet_parchment |         4 |          197.04 | accepted
 DLV-20260930-000046 |   85553 |                 140 | 2026-09-30 07:30:47 | wet_parchment |         3 |          186.73 | accepted
 DLV-20260930-000248 |  215660 |                 166 | 2026-09-30 14:59:13 | cherry        |         3 |          159.89 | received
(10 filas)
```

Ejercicio 9. Rangos de calidad - Análisis de laboratorio

a) Análisis con cup_score mayor o igual a 85

```
SELECT
    T1.analysis_id AS analysis,
    T1.delivery_id AS delivery,
    T1.moisture_pct AS moisture_pct,
    T1.yield_factor AS yield_factor,
    T1.cup_score AS cup_score
FROM buy.quality_analyses AS T1
WHERE T1.analyzed_at >= '2026-09-30'
  AND T1.analyzed_at < '2026-10-01'
  AND T1.cup_score >= 85
ORDER BY cup_score DESC;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.analysis_id AS analysis,
coffee_coop_db->     T1.delivery_id AS delivery,
coffee_coop_db->     T1.moisture_pct AS moisture_pct,
coffee_coop_db->     T1.yield_factor AS yield_factor,
coffee_coop_db->     T1.cup_score AS cup_score
coffee_coop_db-> FROM buy.quality_analyses AS T1
coffee_coop_db-> WHERE T1.analyzed_at >= '2026-09-30'
coffee_coop_db->   AND T1.analyzed_at < '2026-10-01'
coffee_coop_db->   AND T1.cup_score >= 85
coffee_coop_db-> ORDER BY cup_score DESC
coffee_coop_db-> LIMIT 10;
      analysis       |       delivery       | moisture_pct | yield_factor | cup_score
---------------------+----------------------+--------------+--------------+-----------
 QA-20260930-000102 | DLV-20260930-000155 |        12.31 |        94.01 |     87.28
 QA-20260930-000082 | DLV-20260930-000111 |        12.38 |        94.29 |     87.18
 QA-20260930-000137 | DLV-20260930-000295 |        12.35 |        91.07 |     87.14
 QA-20260930-000116 | DLV-20260930-000208 |        11.07 |        93.78 |     87.07
 QA-20260930-000118 | DLV-20260930-000256 |        11.16 |        92.54 |     87.04
 QA-20260930-000030 | DLV-20260930-000089 |        10.70 |        91.53 |     86.31
 QA-20260930-000089 | DLV-20260930-000203 |        11.84 |        91.41 |     86.01
 QA-20260930-000028 | DLV-20260930-000064 |        11.48 |        94.23 |     85.62
 QA-20260930-000110 | DLV-20260930-000182 |        10.64 |        88.18 |     85.25
 QA-20260930-000079 | DLV-20260930-000085 |        11.55 |        93.28 |     85.15
(10 filas)
```

b) Humedad fuera del rango 10%-12% (NOT BETWEEN)

```sql
SELECT
    T1.analysis_id,
    T1.delivery_id,
    T1.analyzed_at,
    T1.moisture_pct,
    T1.yield_factor,
    T1.defects_pct,
    T1.cup_score,
    T1.is_accepted,
    T1.analyst_name,
    T1.observations
FROM buy.quality_analyses AS T1
WHERE T1.analyzed_at >= '2026-09-30'
  AND T1.analyzed_at < '2026-10-01'
  AND T1.moisture_pct NOT BETWEEN 10 AND 12;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.analysis_id, T1.delivery_id, T1.analyzed_at, T1.moisture_pct,
coffee_coop_db->     T1.yield_factor, T1.defects_pct, T1.cup_score, T1.is_accepted,
coffee_coop_db->     T1.analyst_name, T1.observations
coffee_coop_db-> FROM buy.quality_analyses AS T1
coffee_coop_db-> WHERE T1.analyzed_at >= '2026-09-30'
coffee_coop_db->   AND T1.analyzed_at < '2026-10-01'
coffee_coop_db->   AND T1.moisture_pct NOT BETWEEN 10 AND 12
coffee_coop_db-> LIMIT 10;
   analysis_id    |    delivery_id     |     analyzed_at     | moisture_pct | yield_factor | defects_pct | cup_score | is_accepted |   analyst_name   |              observations
------------------+--------------------+---------------------+--------------+--------------+-------------+-----------+-------------+------------------+----------------------------------------
 QA-20260930-000026 | DLV-20260930-000005 | 2026-09-30 10:26:26 |        12.19 |        95.42 |        4.33 |     78.56 | t           | Wilson Galvis    |
 QA-20260930-000018 | DLV-20260930-000006 | 2026-09-30 09:53:20 |        17.24 |       103.31 |        7.88 |     79.98 | f           | Nicol├ís Serrano  | Caf├® con humedad alta, requiere m├ís secado.
 QA-20260930-000017 | DLV-20260930-000012 | 2026-09-30 09:46:31 |        12.15 |        93.62 |        2.04 |           | t           | Natalia Trujillo |
 QA-20260930-000033 | DLV-20260930-000027 | 2026-09-30 10:48:11 |        12.41 |        92.76 |        3.29 |           | t           | Sara S├ínchez    |
 QA-20260930-000013 | DLV-20260930-000038 | 2026-09-30 09:26:29 |        12.28 |        91.77 |        4.74 |     83.22 | t           | Adriana Losada   |
 QA-20260930-000005 | DLV-20260930-000040 | 2026-09-30 07:57:01 |        12.22 |        95.01 |        1.37 |     80.54 | t           | Iv├ín Castro      |
 QA-20260930-000027 | DLV-20260930-000043 | 2026-09-30 10:30:06 |        12.07 |        92.47 |        0.98 |           | t           | Germ├ín Gonz├ílez | Caficultor solicita contramuestra.
 QA-20260930-000102 | DLV-20260930-000155 | 2026-09-30 15:48:14 |        12.31 |        94.01 |        5.26 |     87.28 | t           | Adriana Losada   |
 QA-20260930-000085 | DLV-20260930-000161 | 2026-09-30 14:16:58 |        12.03 |        92.12 |        2.72 |           | t           | Carlos Ospina    | Muestra limpia, color homog├®neo.
 QA-20260930-000068 | DLV-20260930-000165 | 2026-09-30 13:10:38 |        16.13 |       100.22 |       17.29 |           | f           | Germ├ín Pe├▒a     | Muestra tomada por duplicado.
(10 filas)
```

c) No aceptados con observaciones, mayor porcentaje de defectos primero

```sql
SELECT
    T1.analysis_id,
    T1.delivery_id,
    T1.analyzed_at,
    T1.moisture_pct,
    T1.yield_factor,
    T1.defects_pct,
    T1.cup_score,
    T1.is_accepted,
    T1.analyst_name,
    T1.observations
FROM buy.quality_analyses AS T1
WHERE T1.analyzed_at >= '2026-09-30'
  AND T1.analyzed_at < '2026-10-01'
  AND T1.is_accepted = FALSE
  AND T1.observations IS NOT NULL
ORDER BY T1.defects_pct DESC;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.analysis_id,
coffee_coop_db->     T1.delivery_id,
coffee_coop_db->     T1.analyzed_at,
coffee_coop_db->     T1.moisture_pct,
coffee_coop_db->     T1.yield_factor,
coffee_coop_db->     T1.defects_pct,
coffee_coop_db->     T1.cup_score,
coffee_coop_db->     T1.is_accepted,
coffee_coop_db->     T1.analyst_name,
coffee_coop_db->     T1.observations
coffee_coop_db-> FROM buy.quality_analyses AS T1
coffee_coop_db-> WHERE T1.analyzed_at >= '2026-09-30'
coffee_coop_db->   AND T1.analyzed_at < '2026-10-01'
coffee_coop_db->   AND T1.is_accepted = FALSE
coffee_coop_db->   AND T1.observations IS NOT NULL
coffee_coop_db-> ORDER BY T1.defects_pct DESC
coffee_coop_db-> LIMIT 10;
    analysis_id     |     delivery_id     |     analyzed_at     | moisture_pct | yield_factor | defects_pct | cup_score | is_accepted |    analyst_name    |                observations
--------------------+---------------------+---------------------+--------------+--------------+-------------+-----------+-------------+--------------------+---------------------------------------
 QA-20260930-000104 | DLV-20260930-000214 | 2026-09-30 15:55:59 |        17.68 |        97.46 |       19.54 |     82.59 | f           | Carolina Rodr├¡guez | Olor a fermento en la muestra.
 QA-20260930-000068 | DLV-20260930-000165 | 2026-09-30 13:10:38 |        16.13 |       100.22 |       17.29 |           | f           | Germ├¡n Pe├▒a       | Muestra tomada por duplicado.
 QA-20260930-000119 | DLV-20260930-000253 | 2026-09-30 17:15:04 |         8.39 |       106.52 |       15.59 |     84.33 | f           | An├¡bal Pab├│n       | Exceso de impurezas y pasilla.
 QA-20260930-000086 | DLV-20260930-000147 | 2026-09-30 14:23:08 |        15.20 |        96.57 |       13.21 |           | f           | Erika Restrepo     | Olor a fermento en la muestra.
 QA-20260930-000113 | DLV-20260930-000223 | 2026-09-30 16:49:06 |         9.57 |        99.70 |       12.94 |     85.05 | f           | Reinaldo Bermúdez  | Caf├® sobresecado, grano quebradizo.
 QA-20260930-000022 | DLV-20260930-000070 | 2026-09-30 10:11:52 |        12.95 |       103.55 |        9.75 |           | f           | Santiago Rojas     | Caf├® con humedad alta, requiere m├ís secado.
 QA-20260930-000018 | DLV-20260930-000006 | 2026-09-30 09:53:20 |        17.24 |       103.31 |        7.88 |     79.98 | f           | Nicol├ís Serrano    | Caf├® con humedad alta, requiere m├ís secado.
(7 filas)
```

Ejercicio 10. Columnas generadas - Liquidaciones pendientes

a) Liquidaciones del 30 de septiembre de 2026

```sql
SELECT
    T1.settlement_id AS settlement,
    T1.delivery_id AS delivery,
    T1.net_weight_kg AS net_weight_kg,
    T1.price_per_kg_applied AS price_per_kg,
    T1.gross_amount AS gross_cop,
    T1.deductions_amount AS deductions_cop,
    T1.net_amount AS net_cop
FROM pay.settlements AS T1
WHERE T1.settled_at >= '2026-09-30'
  AND T1.settled_at < '2026-10-01';
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.settlement_id AS settlement,
coffee_coop_db->     T1.delivery_id AS delivery,
coffee_coop_db->     T1.net_weight_kg AS net_weight_kg,
coffee_coop_db->     T1.price_per_kg_applied AS price_per_kg,
coffee_coop_db->     T1.gross_amount AS gross_cop,
coffee_coop_db->     T1.deductions_amount AS deductions_cop,
coffee_coop_db->     T1.net_amount AS net_cop
coffee_coop_db-> FROM pay.settlements AS T1
coffee_coop_db-> WHERE T1.settled_at >= '2026-09-30'
coffee_coop_db->   AND T1.settled_at < '2026-10-01'
coffee_coop_db-> LIMIT 10;
     settlement      |      delivery       | net_weight_kg | price_per_kg | gross_cop  | deductions_cop |  net_cop
---------------------+---------------------+---------------+--------------+------------+----------------+------------
 STL-20260930-000025 | DLV-20260927-000082 |        180.06 |     22983.00 | 4138318.98 |      182938.77 | 3955380.21
 STL-20260930-000011 | DLV-20260927-000088 |        211.64 |     22841.00 | 4834069.24 |           0.00 | 4834069.24
 STL-20260930-000018 | DLV-20260927-000097 |        216.32 |     21645.00 | 4682246.40 |           0.00 | 4682246.40
 STL-20260930-000002 | DLV-20260927-000104 |        163.44 |     22367.00 | 3655662.48 |      201901.82 | 3453760.66
 STL-20260930-000035 | DLV-20260927-000128 |        237.30 |     22840.00 | 5419932.00 |           0.00 | 5419932.00
 STL-20260930-000027 | DLV-20260927-000144 |        139.52 |     21607.00 | 3014608.64 |           0.00 | 3014608.64
 STL-20260930-000003 | DLV-20260927-000169 |        338.42 |     20409.00 | 6906813.78 |      344352.08 | 6562461.70
 STL-20260930-000007 | DLV-20260927-000177 |        226.45 |     20916.00 | 4736428.20 |      167394.38 | 4569033.82
 STL-20260930-000030 | DLV-20260927-000197 |         39.43 |     23273.00 |  917654.39 |           0.00 |  917654.39
 STL-20260930-000005 | DLV-20260927-000198 |        158.94 |     21608.00 | 3434375.52 |           0.00 | 3434375.52
(10 filas)


coffee_coop_db=>
```

b) Liquidaciones en estado pending, mayor valor neto primero

```sql
SELECT
    T1.settlement_id,
    T1.delivery_id,
    T1.daily_price_id,
    T1.net_weight_kg,
    T1.price_per_kg_applied,
    T1.deductions_amount,
    T1.gross_amount,
    T1.net_amount,
    T1.settled_at,
    T1.status
FROM pay.settlements AS T1
WHERE T1.settled_at >= '2026-09-30'
  AND T1.settled_at < '2026-10-01'
  AND T1.status = 'pending'
ORDER BY T1.net_amount DESC;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.settlement_id, T1.delivery_id, T1.daily_price_id, T1.net_weight_kg,
coffee_coop_db->     T1.price_per_kg_applied, T1.deductions_amount, T1.gross_amount,
coffee_coop_db->     T1.net_amount, T1.settled_at, T1.status
coffee_coop_db-> FROM pay.settlements AS T1
coffee_coop_db-> WHERE T1.settled_at >= '2026-09-30'
coffee_coop_db->   AND T1.settled_at < '2026-10-01'
coffee_coop_db->   AND T1.status = 'pending'
coffee_coop_db-> ORDER BY T1.net_amount DESC
coffee_coop_db-> LIMIT 10;
    settlement_id    |     delivery_id     | daily_price_id | net_weight_kg | price_per_kg_applied | deductions_amount | gross_amount | net_amount |     settled_at      | status
---------------------+---------------------+----------------+---------------+----------------------+-------------------+--------------+------------+---------------------+---------
 STL-20260930-000062 | DLV-20260929-000235 |         311929 |        283.06 |             21803.00 |              0.00 |   6171557.18 | 6171557.18 | 2026-09-30 21:43:06 | pending
 STL-20260930-000013 | DLV-20260928-000004 |         311759 |        236.95 |             22339.00 |         205370.50 |   5293226.05 | 5087855.55 | 2026-09-30 04:28:06 | pending
 STL-20260930-000069 | DLV-20260930-000204 |         312055 |        231.21 |             22692.00 |         232910.66 |   5246617.32 | 5013706.66 | 2026-09-30 23:55:18 | pending
 STL-20260930-000021 | DLV-20260928-000105 |         311675 |        219.83 |             22347.00 |              0.00 |   4912541.01 | 4912541.01 | 2026-09-30 07:04:02 | pending
 STL-20260930-000011 | DLV-20260927-000088 |         311441 |        211.64 |             22841.00 |              0.00 |   4834069.24 | 4834069.24 | 2026-09-30 03:43:39 | pending
 STL-20260930-000018 | DLV-20260927-000097 |         311405 |        216.32 |             21645.00 |              0.00 |   4682246.40 | 4682246.40 | 2026-09-30 06:29:26 | pending
 STL-20260930-000007 | DLV-20260927-000177 |         311539 |        226.45 |             20916.00 |         167394.38 |   4736428.20 | 4569033.82 | 2026-09-30 02:14:11 | pending
 STL-20260930-000068 | DLV-20260928-000311 |         311662 |        192.15 |             22592.00 |              0.00 |   4341052.80 | 4341052.80 | 2026-09-30 23:37:11 | pending
 STL-20260930-000033 | DLV-20260929-000007 |         311880 |        183.79 |             22164.00 |              0.00 |   4073521.56 | 4073521.56 | 2026-09-30 12:15:08 | pending
 STL-20260930-000016 | DLV-20260928-000207 |         311707 |        180.13 |             21779.00 |              0.00 |   3923051.27 | 3923051.27 | 2026-09-30 06:05:30 | pending
(10 filas)
```

c) Liquidaciones sin deducciones y columna de verificación

```sql
SELECT
    T1.settlement_id,
```

```sql
    T1.delivery_id,
    T1.daily_price_id,
    T1.net_weight_kg,
    T1.price_per_kg_applied,
    T1.deductions_amount,
    T1.gross_amount,
    T1.net_amount,
    T1.settled_at,
    T1.status,
    T1.net_weight_kg * T1.price_per_kg_applied AS check_gross_cop
FROM pay.settlements AS T1
WHERE T1.settled_at >= '2026-09-30'
  AND T1.settled_at < '2026-10-01'
  AND T1.deductions_amount = 0;
```

```
coffee_coop_db=> SELECT
coffee_coop_db->     T1.settlement_id, T1.delivery_id, T1.daily_price_id, T1.net_weight_kg,
coffee_coop_db->     T1.price_per_kg_applied, T1.deductions_amount, T1.gross_amount,
coffee_coop_db->     T1.net_amount, T1.settled_at, T1.status,
coffee_coop_db->     T1.net_weight_kg * T1.price_per_kg_applied AS check_gross_cop
coffee_coop_db-> FROM pay.settlements AS T1
coffee_coop_db-> WHERE T1.settled_at >= '2026-09-30'
coffee_coop_db->   AND T1.settled_at < '2026-10-01'
coffee_coop_db->   AND T1.deductions_amount = 0
coffee_coop_db-> LIMIT 10;
    settlement_id     |     delivery_id     | daily_price_id | net_weight_kg | price_per_kg_applied | deductions_amount | gross_amount | net_amount |      settled_at      | status  | check_gross_cop
----------------------+---------------------+----------------+---------------+----------------------+-------------------+--------------+------------+----------------------+---------+-----------------
 STL-20260930-000011 | DLV-20260927-000088 |         311441 |        211.64 |             22841.00 |              0.00 |   4834069.24 | 4834069.24 | 2026-09-30 03:43:39 | pending |    4834069.2400
 STL-20260930-000018 | DLV-20260927-000097 |         311485 |        216.32 |             21645.00 |              0.00 |   4682246.40 | 4682246.40 | 2026-09-30 06:29:26 | pending |    4682246.4000
 STL-20260930-000035 | DLV-20260927-000128 |         311569 |        237.30 |             22840.00 |              0.00 |   5419932.00 | 5419932.00 | 2026-09-30 12:24:40 | paid    |    5419932.0000
 STL-20260930-000027 | DLV-20260927-000144 |         311486 |        139.52 |             21607.00 |              0.00 |   3014608.64 | 3014608.64 | 2026-09-30 10:24:19 | pending |    3014608.6400
 STL-20260930-000030 | DLV-20260927-000197 |         311432 |         39.43 |             23273.00 |              0.00 |    917654.39 |  917654.39 | 2026-09-30 10:50:17 | pending |     917654.3900
 STL-20260930-000005 | DLV-20260927-000198 |         311520 |        158.94 |             21608.00 |              0.00 |   3434375.52 | 3434375.52 | 2026-09-30 01:39:34 | paid    |    3434375.5200
 STL-20260930-000042 | DLV-20260928-000048 |         311743 |        300.18 |             23455.00 |              0.00 |   7040721.90 | 7040721.90 | 2026-09-30 14:42:06 | paid    |    7040721.9000
 STL-20260930-000006 | DLV-20260928-000058 |         311636 |        145.01 |             22847.00 |              0.00 |   3313043.47 | 3313043.47 | 2026-09-30 01:42:33 | pending |    3313043.4700
 STL-20260930-000019 | DLV-20260928-000075 |         311653 |        134.15 |             22808.00 |              0.00 |   3059693.20 | 3059693.20 | 2026-09-30 06:39:41 | pending |    3059693.2000
 STL-20260930-000008 | DLV-20260928-000093 |         311739 |        152.95 |             22477.00 |              0.00 |   3437857.15 | 3437857.15 | 2026-09-30 02:28:33 | paid    |    3437857.1500
(10 filas)
```

Explicación

(Escribe aquí, con tus propias palabras, por qué `check_gross_cop` coincide con `gross_amount`.)